

MakerSpace - Task 2

Teach the Class: Arduino Basics

Objective

In Task 1, you researched what Arduino is, how it is used, and the tools programmers use to work with it. Now you will go one step further by becoming the teacher. Your goal is to help the class understand one important Arduino concept before we begin working with the board.

Your Task

You will be assigned one Arduino topic. Research your topic and prepare a **short presentation** to teach the class. Your presentation should be approximately **5 minutes** and should focus on the most important basic ideas your classmates need to understand.

- **Prepare a presentation** with about 3-5 clear slides.
- Explain the topic **in your own words**.
- Teach the most important vocabulary or concepts.
- Include at least **one visual, example, or demonstration**.
- Explain **why the topic is important** when using Arduino.
- End with **one quick question** to check the class's understanding.
- Do not read your slides word for word. Use them to support what you are teaching.

Research Rules - Read First, Then Create

This assignment is designed to practice real research and reading. AI tools may not be used to find, summarize, or write the information for this presentation.

- Use **at least two reliable sources**. Official Arduino pages, documentation, educational websites, and teacher-provided links are good choices.
- **Open and read** the sources yourself. Do not use AI-generated search summaries as a source.
- Take short notes **in your own words** before you begin making your slides.
- Add a final **Sources** slide with the title or website name and the URL for each source you used.
- Be ready for a follow-up question. You should be able to explain your topic without depending on the exact wording on your slides.

Arduino Topics

Each student will be assigned one topic.

1. What Is Arduino?

Explain what Arduino is, what it can control, and give a few examples of things people build with it.

2. Arduino Hardware Overview

Introduce the main parts of an Arduino board: microcontroller, digital pins, analog pins, power pins, USB connection, reset button, and other important parts.

3. Install and Set Up the Arduino IDE

Show how to download and install the Arduino IDE. Explain how to connect an Arduino board, select the correct board, and select the correct port.

4. Arduino IDE and Sketch Overview

Explain the main parts of the Arduino IDE and what an Arduino sketch is. Introduce **setup()** and **loop()** and explain how a sketch is uploaded to the board.

5. Arduino Syntax Basics

Explain basic rules used when writing Arduino code: braces { }, semicolons ;, comments, functions, and why syntax must be written correctly.

6. Understanding Variables

Explain what a variable is, why programmers use variables, and show a few simple Arduino examples using variables such as **int** and **bool**.

7. Understanding Inputs and Outputs

Explain what an **input** and an **output** are in Arduino. Give examples of inputs such as buttons or sensors and outputs such as LEDs, buzzers, or motors. Explain how Arduino can receive information and then control something.

8. How to Blink an LED

Show how the classic Blink program works. Explain what the code does and how Arduino turns an LED on and off. Connect the example to concepts taught in the earlier mini lessons.

While Your Classmates Are Teaching

Write down at least one important idea from each mini lesson. You will use these notes in our next Arduino activity, so listen for the key concept in every presentation.

Reminder

The goal is not to give a perfect or advanced lecture. Your job is to make one Arduino idea **clear, simple, and useful** for the rest of the class.